

13.2 Jensen equation

The equation resulting on replacing in the Jensen inequality (5.3.1) the sign of inequality by that of equality

$$f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2} \quad (13.2.1)$$

is known as the *Jensen equation* (cf. Aczél [5]). We will consider equation (13.2.1) for functions $f : D \rightarrow \mathbb{R}$, where $D \subset \mathbb{R}^N$ is a convex set. If D were also open, then f satisfying (13.2.1) would be convex, and thus all the results established in the previous chapters of this book for convex functions would apply. But most of these results become invalid when the set D is no longer open. For example, let $N = 1$, $D = [-1, 1]$, and let $f : D \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x^2 & \text{if } x \in (-1, 1), \\ 2 & \text{if } |x| = 1. \end{cases}$$

Such an f satisfies inequality (5.3.1) in $D = [-1, 1]$, is measurable and bounded in D , but is discontinuous at $x = 1$ and $x = -1$. So we treat equation (13.2.1) in a more general setting.

Lemma 13.2.1. *Let $D \subset \mathbb{R}^N$ be a convex set such that $0 \in D$, and let $f : D \rightarrow \mathbb{R}$ be a solution of equation (13.2.1) such that*

$$f(0) = 0. \quad (13.2.2)$$

Then, for every $x \in D$ and $n \in \mathbb{N}$,

$$f\left(\frac{x}{2^n}\right) = \frac{1}{2^n} f(x). \quad (13.2.3)$$

Proof. Take an $x \in D$. Since D is convex, $\frac{1}{2}(x+0) \in D$, and by (13.2.1) and (13.2.2)

$$\begin{aligned} f\left(\frac{x}{2}\right) &= f\left(\frac{x+0}{2}\right) \\ &= \frac{f(x) + f(0)}{2} = \frac{f(x)}{2}. \end{aligned} \quad (13.2.4)$$

Thus (13.2.3) holds for $n = 1$. Assuming it true for an $n \in \mathbb{N}$, we have

$$\frac{x}{2^{n+1}} = \frac{1}{2^{n+1}}x + \left(1 - \frac{1}{2^{n+1}}\right)0 \in D,$$

and by (13.2.3) (for n) and (13.2.4)

$$f\left(\frac{x}{2^{n+1}}\right) = \frac{1}{2^n} f\left(\frac{x}{2}\right) = \frac{1}{2^{n+1}} f(x).$$

Induction completes the proof. □

Lemma 13.2.2. *Let $D \subset \mathbb{R}^N$ be a convex set such that $\text{int } D \neq \emptyset$, and let $f : D \rightarrow \mathbb{R}$ be a solution of equation (13.2.1). Fix an $x_0 \in \text{int } D$, and define the function $f_0 : (D - x_0) \rightarrow \mathbb{R}$ by*

$$f_0(x) = f(x_0 + x) - f(x_0). \quad (13.2.5)$$

Then there exists a unique function $f_1 : \mathbb{R}^N \rightarrow \mathbb{R}$ satisfying equation (13.2.1) in $\mathbb{R}^N$ and such that

$$f_1(x) = f_0(x) \text{ for } x \in D - x_0. \quad (13.2.6)$$

Proof. Function (13.2.5) is defined for $x \in D - x_0$. First we verify that f_0 satisfies equation (13.2.1) in $D - x_0$. For every $x, y \in D - x_0$, we have $x_0 + x, x_0 + y \in D$, and by (13.2.5) and (13.2.1)

$$\begin{aligned} f_0\left(\frac{x+y}{2}\right) &= f\left(x_0 + \frac{x+y}{2}\right) - f(x_0) = f\left(\frac{x_0 + x + x_0 + y}{2}\right) - f(x_0) \\ &= \frac{1}{2}f(x_0 + x) + \frac{1}{2}f(x_0 + y) - f(x_0) \\ &= \frac{1}{2}[f(x_0 + x) - f(x_0)] + \frac{1}{2}[f(x_0 + y) - f(x_0)] \\ &= \frac{1}{2}[f_0(x) + f_0(y)]. \end{aligned}$$

Also, it is easily seen that $0 \in D - x_0$ and by (13.2.5)

$$f_0(0) = 0. \quad (13.2.7)$$

Now put $D_0 = D - x_0$ and

$$D_n = 2^n D_0, \quad n \in \mathbb{N}.$$

If $x \in D_n$, then $\frac{x}{2^n} \in D_0$. D_0 is convex, just like D , and $0 \in D_0$, whence $\frac{x}{2^{n+1}} = \frac{1}{2} \left[\frac{x}{2^n} + 0 \right] \in D_0$, and $x \in 2^{n+1} D_0 = D_{n+1}$. Thus

$$D_n \subset D_{n+1}, \quad n \in \mathbb{N} \cup \{0\}. \quad (13.2.8)$$

Also, $0 \in \text{int } D_0$, since $x_0 \in \text{int } D$. For every $x \in \mathbb{R}^N$ we have $\lim_{n \rightarrow \infty} x/2^n = 0$, whence it follows that there exists an $n \in \mathbb{N} \cup \{0\}$ such that $x/2^n \in D_0$, whence $x \in D_n$. Hence

$$\bigcup_{n=0}^{\infty} D_n = \mathbb{R}^N. \quad (13.2.9)$$

Define the function $f_1 : \mathbb{R}^N \rightarrow \mathbb{R}$ as follows:

$$f_1(x) = 2^n f_0\left(\frac{x}{2^n}\right) \text{ if } x \in D_n, n \in \mathbb{N} \cup \{0\}. \quad (13.2.10)$$

First we must check whether definition (13.2.10) is correct. If $x \in D_n = 2^n D_0$, then $x/2^n \in D_0 = D - x_0$, and the expression $f_0(x/2^n)$ is meaningful. If $x \in D_n$ and

$x \in D_m$ for $n \neq m$, say $n < m$, then by (13.2.7) and Lemma 13.2.1

$$2^m f_0 \left(\frac{x}{2^m} \right) = 2^m \frac{1}{2^{m-n}} f_0 \left(\frac{x}{2^n} \right) = 2^n f_0 \left(\frac{x}{2^n} \right),$$

and thus (13.2.10) is unambiguous. By (13.2.9) the function f_1 is defined in the whole of $\mathbb{R}^N$. We must verify that it satisfies equation (13.2.1) in $\mathbb{R}^N$.

Take arbitrary $x, y \in \mathbb{R}^N$. By (13.2.9) and (13.2.8) there exists an $n \in \mathbb{N} \cup \{0\}$ such that $x, y \in D_n$. Hence $x/2^n, y/2^n \in D_0 = D - x_0$, and $\frac{x+y}{2^{n+1}} = \frac{1}{2} \left(\frac{x}{2^n} + \frac{y}{2^n} \right) \in D_0$, whence $\frac{x+y}{2} \in D_n$. Now,

$$\begin{aligned} f_1 \left(\frac{x+y}{2} \right) &= 2^n f_0 \left(\frac{x+y}{2^{n+1}} \right) \\ &= 2^n \left[\frac{1}{2} f_0 \left(\frac{x}{2^n} \right) + \frac{1}{2} f_0 \left(\frac{y}{2^n} \right) \right] \\ &= \frac{1}{2} \left[2^n f_0 \left(\frac{x}{2^n} \right) + 2^n f_0 \left(\frac{y}{2^n} \right) \right] = \frac{f_1(x) + f_1(y)}{2}. \end{aligned}$$

Relation (13.2.6) results from (13.2.10) for $n = 0$.

To prove the uniqueness, suppose that a function $f_2 : \mathbb{R}^N \rightarrow \mathbb{R}$ satisfies equation (13.2.1) in $\mathbb{R}^N$ and fulfils the condition

$$f_2(x) = f_0(x) \quad \text{for } x \in D - x_0. \quad (13.2.11)$$

By (13.2.11) and (13.2.7) $f_2(0) = f_0(0) = 0$, and hence, by Lemma 13.2.1, $f_2 \left(\frac{x}{2^n} \right) = \frac{1}{2^n} f_2(x)$, $x \in \mathbb{R}^N$, $n \in \mathbb{N} \cup \{0\}$. Take an arbitrary $x \in \mathbb{R}^N$. By (13.2.9) there exists an $n \in \mathbb{N} \cup \{0\}$ such that $x \in D_n = 2^n D_0$, whence $x/2^n \in D_0 = D - x_0$. Thus we have by (13.2.11) and (13.2.10)

$$f_2(x) = 2^n f_2 \left(\frac{x}{2^n} \right) = 2^n f_0 \left(\frac{x}{2^n} \right) = f_1(x).$$

Consequently $f_2 = f_1$ in $\mathbb{R}^N$. □

Lemma 13.2.3. *Let a function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ satisfy equation (13.2.1) and relation (13.2.2). Then f is additive.*

Proof. We have by Lemma 13.2.1 for arbitrary $x, y \in \mathbb{R}^N$

$$f(x+y) = 2f \left(\frac{x+y}{2} \right) = 2 \frac{f(x) + f(y)}{2} = f(x) + f(y),$$

i.e., f is additive. □

Theorem 13.2.1. *Let $D \subset \mathbb{R}^N$ be a convex set such that $\text{int } D \neq \emptyset$, and let $f : D \rightarrow \mathbb{R}$ be a solution of equation (13.2.1). Then there exist an additive function $g : \mathbb{R}^N \rightarrow \mathbb{R}$ and a constant $a \in \mathbb{R}$ such that*

$$f(x) = g(x) + a \quad \text{for } x \in D. \quad (13.2.12)$$

Proof. Fix an $x_0 \in \text{int } D$ and define the function $f_0 : (D - x_0) \rightarrow \mathbb{R}$ by (13.2.5). By Lemma 13.2.2 there exists a function $f_1 : \mathbb{R}^N \rightarrow \mathbb{R}$ satisfying equation (13.2.1) and condition (13.2.6). Hence by (13.2.7) $f_1(0) = f_0(0) = 0$. By Lemma 13.2.3 f_1 is additive.

For arbitrary $x \in D$ we have $x - x_0 \in D - x_0$, whence by (13.2.5) and (13.2.6)

$$f(x) = f(x_0 + (x - x_0)) = f_0(x - x_0) + f(x_0) = f_1(x - x_0) + f(x_0).$$

Since f_1 is additive, we get hence

$$f(x) = f_1(x) - f_1(x_0) + f(x_0). \quad (13.2.13)$$

Put $g = f_1$, $a = f(x_0) - f_1(x_0)$. Relation (13.2.12) results now from (13.2.13). $\square$

It is easily seen that every function of form (13.2.12), where $g : \mathbb{R}^N \rightarrow \mathbb{R}$ is additive, actually satisfies equation (13.2.1).

Theorem 13.2.2. *Let $D \subset \mathbb{R}^N$ be a convex set such that $\text{int } D \neq \emptyset$. A function $f : D \rightarrow \mathbb{R}$ is a continuous solution of equation (13.2.1) if and only if*

$$f(x) = cx + a, \quad x \in D, \quad (13.2.14)$$

with certain constants $c \in \mathbb{R}^N$, $a \in \mathbb{R}$.

Proof. By Theorem 13.2.1 we have (13.2.12) with an additive $g : \mathbb{R}^N \rightarrow \mathbb{R}$. We have $g = f - a$ in D , and since f is continuous, also g is continuous in $\text{int } D$ and hence (Theorem 6.4.3) in $\mathbb{R}^N$. By Theorem 5.5.2 $g(x) = cx$ with a certain constant $c \in \mathbb{R}^N$. Thus (13.2.14) results from (13.2.12).

It is obvious that every function of form (13.2.14) is a continuous solution of (13.2.1). $\square$

Even if D is convex and open, then, using the fact that f fulfilling (13.2.1) is convex, we can infer on its continuity only from the boundedness above on a suitable set. If D is not open, then we cannot do even this, as it has been pointed out at the beginning of this section. But from Theorem 13.2.1 we obtain the following result.

Theorem 13.2.3. *Let $D \subset \mathbb{R}^N$ be a convex set such that $\text{int } D \neq \emptyset$, and let $f : D \rightarrow \mathbb{R}$ be a solution of equation (13.2.1). If f is measurable, or is bounded above, or below, on a set $T \in \mathfrak{A}$, then f is continuous.*

Proof. By Theorem 13.2.1 we have (13.2.12), and if f is measurable, then so is also $g \mid \text{int } D$. The function $g \mid \text{int } D$ is convex (Theorem 5.1.5), and hence, by Theorem 9.4.1 is continuous. By Theorem 6.4.3 g is continuous, and thus, by (13.2.12), f is continuous.

If f is bounded above, or below, on a set $T \subset D$, then so is also g . Since $T \in \mathfrak{A}$, this implies that g is continuous (if g is bounded below on T , then $-g$ is bounded above on T), and hence, by (13.2.12), f is continuous. $\square$